

Assignment "2"

Topic : Pseudo-code

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Question : 1

In a right angle triangle, the square of the length of one side is equal to the sum of the squares of the length of the other two sides. Write a pseudo-code that takes in length of the three sides of a triangle, and output whether the triangle is right or not.

SOLUTION :

$sq\ x_1$, $sq\ x_2$, $sq\ x_3$

$sq\ x_1 = 0$

$sq\ x_2 = 0$

$sq\ x_3 = 0$

input x_1 , x_2 , x_3

$sq\ x_1 = x_1 * x_1$

$sq\ x_3 = x_3 * x_3$

$sq\ x_2 = x_2 * x_2$

if $(sq\ x_1 = sq\ x_2 + sq\ x_3$

OR

$sq\ x_2 = sq\ x_3 + sq\ x_1$

OR

$sq\ x_3 = sq\ x_1 + sq\ x_2$)

Print "Right Triangle"

else

Print "Not a Right Triangle"

Question 02

write a pseudo-code that takes in a number as input and prints whether it is negative, positive or zero.

Solution :

input Number

if (Number > 0)

Print "Positive"

else if (Number < 0)

Print "Negative"

else

Print "zero"

Question 03

Write a Pseudo-code that Takes in as input the number of units used by the consumer, and print the total cost of electricity according to the given conditions;

- For First 110 units, the rate is 0.50/unit
- For Next 90 units, the rate is 0.80/unit
- For above 200 units, the rate gets raised to Rs 1.50/unit

Solution :

electricity Bill

electricity Bill = 0

input Nunit

if (Unit $\leq$ 110)

electricity Bill = Unit * 0.50

else if (Unit > 110 AND Unit $\leq$ 200)

electricity Bill = $110 * 0.50 + (\text{Unit} - 110) * 0.80$

else

electricity Bill = $110 * 0.50 + 90 * 0.80 + (\text{Unit} - 200) * 1.50$

Print electricity Bill.
